

Christopher Jacobs

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EDUCATION

Clemson University

Doctorate in Computer Science and Visual Computing

Charleston, S.C.

August 2024 - May 2028

Clemson University

Master of Science in Digital Production Arts; GPA: 3.95

Charleston, S.C.

August 2022 - May 2024

Francis Marion University

Bachelor of Science in Computer Science; GPA: 3.96

Florence, S.C.

August 2018 - May 2022

SKILLS SUMMARY

- **Languages:** Python, C/C++ Java, SQL
- **Frameworks:** PyTorch, NumPy, OpenCV, Matplotlib, PyQt 5, PySide 6, OpenGL, CUDA,
- **Tools:** Unreal Engine 4/5, Autodesk Maya, Houdini, Nuke, Perforce
- **Research Areas:** Real-Time Rendering, Neural Rendering, Computer Vision, 3D Reconstruction
- **Technical Skills:** Version Control (Git), Linux/Windows, Bash Scripting, Performance Optimization

WORK EXPERIENCE

RESEARCH ASSISTANT | CLEMSON UNIVERSITY

January 2024 - Present

- Engineered a real-time middleware system between Project Chrono and Unreal Engine (C++/Python) enabling high-fidelity physical simulations for virtual prototyping and visualization.
- Designed a hybrid physically-informed neural network (PINN) framework that could accelerate vehicle simulation in Unreal by reducing computational overhead while maintaining physical accuracy.

ASSISTANT MANAGER | AMERICAN EAGLE OUTFITTERS

August 2021 - May 2022

- Lead and optimize selling team and merchandising team performance to increase store revenue (+3,583 from 2021 LY) and conversion rates (+2.3% from 2021 LY).
- Train and onboard new associates on company policies, store procedures, and product knowledge while maintaining a welcoming and inclusive environment for both guests and brand ambassadors.

PROJECTS

Physically Based Ray Marcher for Visual Effects

Implemented a physically accurate ray marcher using C++ for volumetric effects such as spline wisps, noise clouds, and terrain synthesis, all of which mimic film-quality VFX pipelines.

Feature Detection with U-Net Architecture

Analyzing spatial feature detection through the utilization of a U-Net + ResNet model on objects and faces to aid in 3D reconstruction.

Customizable Middleware for Physics Simulations

Designed a modular C++/Python middleware framework that allows for seamless interchange of visualization and physics for simulation testing.